X (Twitter) Sentiment Analysis Report

1. Project Overview

This project performs sentiment analysis on X (Twitter) data using VADER (Valence Aware Dictionary and sEntiment Reasoner).

Total Tweets Analysed : 162,975

Unique Categories : 3

Tools Used : Python, VADER, NLTK, Pandas, Matplotlib

2. Sentiment Distribution

Positive Tweets : 78,194 (48.0%)

Neutral Tweets : 34,838 (21.4%)

Negative Tweets : 49,943 (30.6%)

Average Compound Score : 0.1019

Median Compound Score : 0.0000

Overall Sentiment : Positive

3. Category Insights

Most Positive Category : 1.0 (Score: 0.3058)

Most Negative Category : -1.0 (Score: -0.1894)

4. Key Findings

1. The dominant sentiment is 'Positive' (48.0% of all tweets).
2. Top Positive Words : modi, india, like, bjp, people
3. Top Negative Words : modi, india, people, congress, bjp
4. '1.0' drives the most positive discussions.
5. '-1.0' generates the most negative reactions.

5. Sentiment Dashboard

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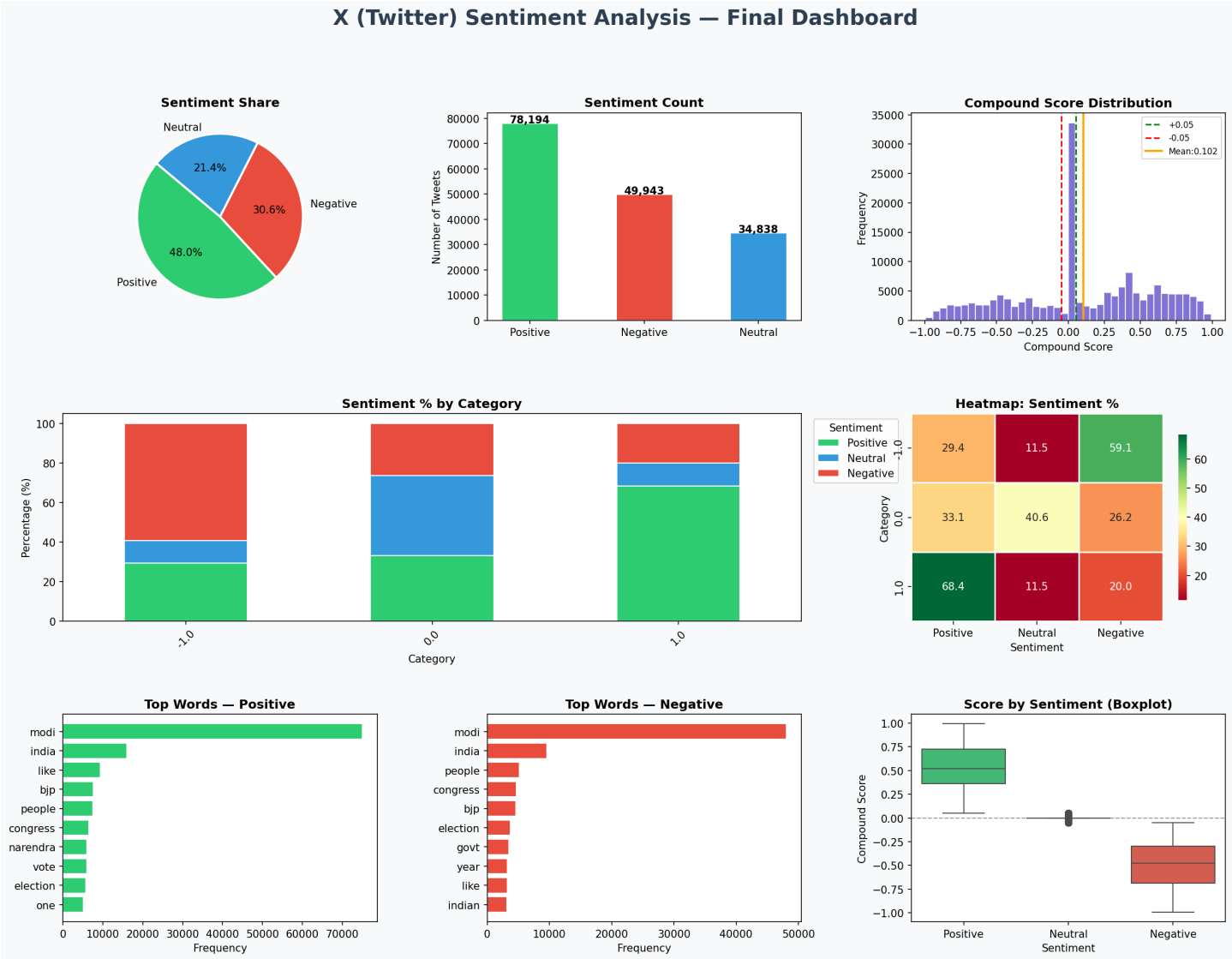

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6. Conclusion

This project successfully applied NLP-based sentiment analysis on X (Twitter) data. Using VADER, tweets were classified into Positive, Negative, and Neutral categories. The analysis revealed meaningful patterns in public opinion across different categories. These findings can assist businesses and researchers in understanding public discourse on social media platforms.

Limitations:

- VADER may misclassify sarcasm or irony
- Dataset may not represent all demographics
- Slang and abbreviations may affect accuracy